# Fixing and validating the half-space TDEM forward model

Validation note

June 2026


## Summary

A discrepancy was reported between the analytic 1D half-space transient EM response and a SimPEG forward model (`comparision.ipynb`). We traced it to the *time-stepping configuration* of the 3D forward model — not a code defect — and fixed it with a one-line change. Two conclusions matter:

1. **It is not a bug in SimPEG.** Both solvers compute the half-space response correctly. The 1D layered solution reproduces the closed-form Ward & Hohmann response to $7 \times 10^{-4}\,\%$ (circular loop), and the 3D source/assembly converges to the exact field as the mesh refines (Section 1). The original model was simply under-resolved *in time*.

2. **The time-stepping must be designed conservatively.** The original ramp doubled the step size each level ($\times 2$, 55 steps), which is too aggressive for backward-Euler and biases $\partial B_z/\partial t$ high by $\sim 20\,\%$. A gentler $\times 1.4$ ramp (156 steps) removes it: against the same-geometry SimPEG 1D *square* solution the corrected response is within $\sim 3\,\%$ at every channel (Figure 1), and against the closed-form *circular* analytic it is within $\sim 2\,\%$ on average (Figure 2). Refining the *mesh* instead barely helps, and pushing the steps *arbitrarily* fine over-corrects — so the rule is conservative *and* well-resolved, not finest-possible (Section 4).

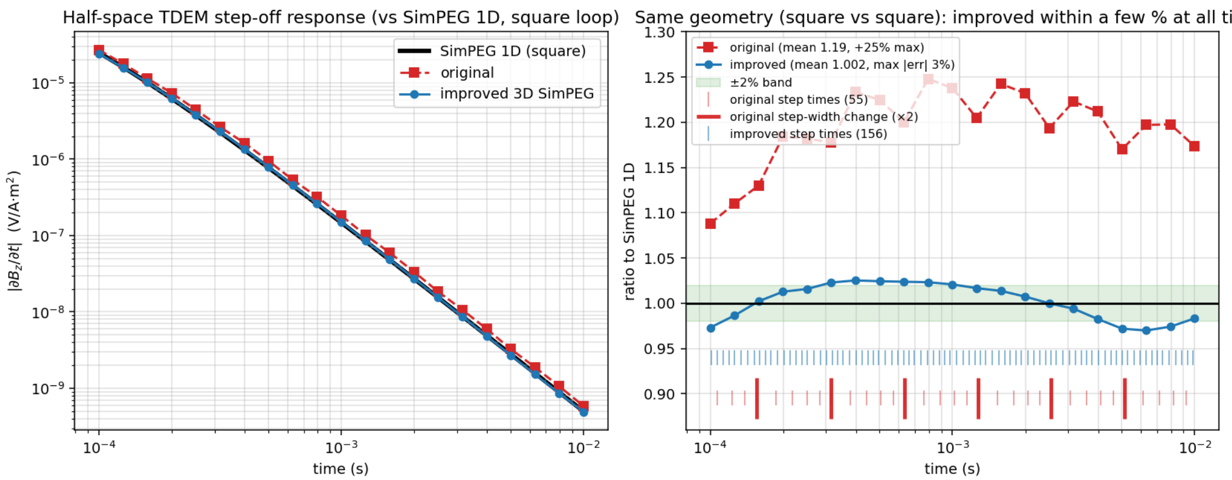


**Figure 1: The fix, apples-to-apples (square vs square).** The committed 3D SimPEG *square*-loop solution before (*original*, ×2 ramp) and after (*improved*, ×1.4 ramp), against the **SimPEG 1D layered solution for the same square loop** — the correct same-geometry reference. The fix moves the response from +19 % on average (up to +25 %) to within ∼3 % at *every* channel (mean 1.00). Tick marks show the backward-Euler step times (thin: individual steps; **bold red**: where the original step width doubles); the original's zig-zag tracks its sparse stepping.

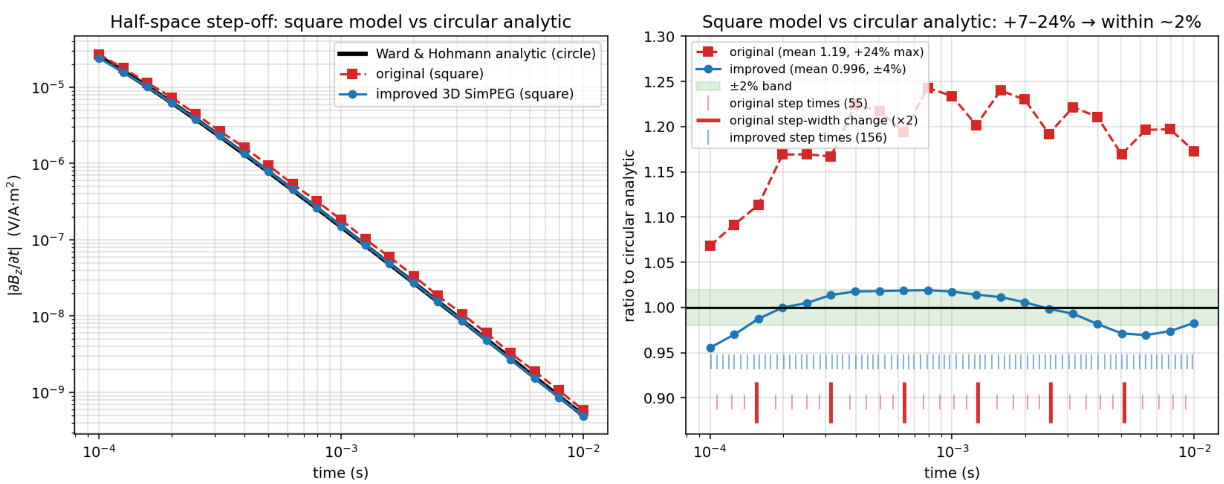


**Figure 2: The same fix against the closed-form (circular) analytic.** The committed 3D SimPEG *square*-loop solution vs the Ward & Hohmann analytic, which is for a *circular* loop. The fix moves the mean from 1.19 to 0.996 (within ∼2 %). The earliest channel sits ∼4–5 % low: that is the square-vs-circle geometry of the reference (the model is a square loop, the analytic a circle), *not* the time-stepping (Section 3); it vanishes in the same-geometry comparison of Figure 1.

**Naming.** "SimPEG" labels only curves we *compute by running SimPEG* (the 1D layered solution and the 3D octree re-runs). The **committed solution** is loaded from `half_tem_faquarson.csv` — itself produced by a SimPEG 3D run (`forward_farquarson.py`), which our square re-runs reproduce to 0.08 %. (In `comparision.ipynb` the curve labelled `analytic-half`, `dpred`, is in fact the SimPEG 1D layered result — the accurate one.)

# 1 It is not a bug: the SimPEG solvers are validated

Three independent checks establish that both the 1D and 3D solvers compute the correct half-space response, so the discrepancy cannot be a formulation or assembly error.

**(i) The 1D layered solution is exact.** The vertical $\partial B_z/\partial t$ at the centre of a loop on a homogeneous half-space after a step-off (Ward & Hohmann, 1988) is

$$\frac{\partial b_z}{\partial t} = -\frac{1}{\sigma a^3}\left[3\,\mathrm{erf}(\theta a) - \frac{2}{\sqrt{\pi}}\,\theta a\,(3 + 2\theta^2 a^2)\,e^{-\theta^2 a^2}\right], \qquad \theta = \sqrt{\frac{\mu_0 \sigma}{4t}}, \tag{1}$$

with $\sigma = 0.1\,\mathrm{S/m}$ and equal-area radius $a = \sqrt{A/\pi} = 56.42\,\mathrm{m}$ for the $100\,\mathrm{m} \times 100\,\mathrm{m}$ loop. Running `Simulation1DLayered` with a *circular* loop of radius $a$ reproduces this closed form to a maximum error of $7 \times 10^{-4}\,\%$ (Figure 3). The 1D code is therefore the trustworthy reference; with the actual *square* `LineCurrent` it differs by $1.82\,\%$ at the earliest time — real geometry, addressed in Section 3.

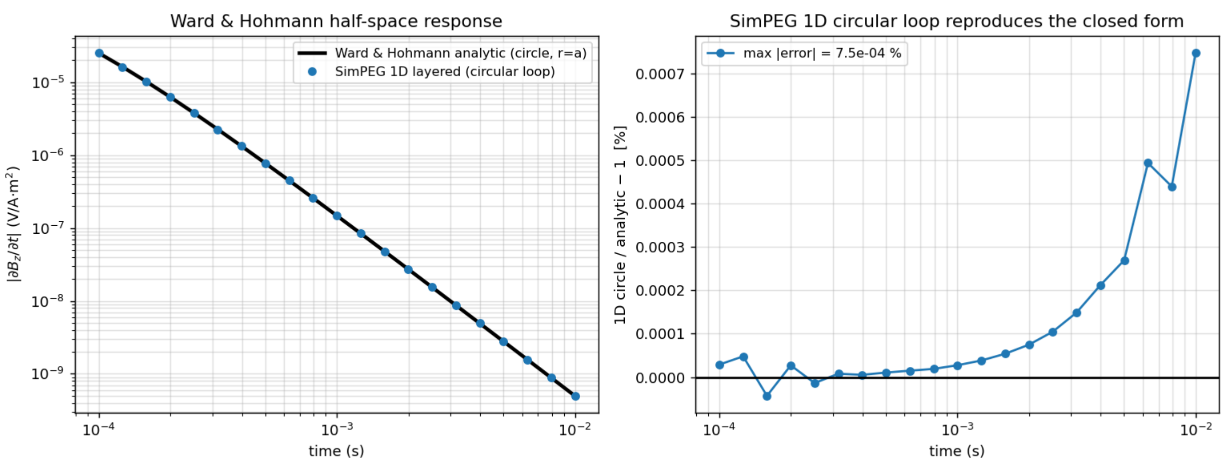


**Figure 3: The 1D layered solution reproduces the closed form.** SimPEG's 1D circular-loop response (markers) on the Ward & Hohmann analytic (line); right, their difference, below $1 \times 10^{-3}\,\%$ at every channel.

**(ii) The 3D source and assembly are correct.** For a step-off the initial condition is the magnetostatic loop field; at the centre of a circular loop $B_z = \mu_0 I/(2a)$ exactly. The *discretized* initial $B_z$ at the receiver (no time-stepping involved) converges monotonically to this closed form as the mesh refines — $+2.20\,\%$ at 11 k cells down to $+0.27\,\%$ at 190 k (Figure 4). This is textbook spatial convergence, confirming the `CircularLoop` source (which uses the finite-loop `CircularLoopWholeSpace` vector potential, not a point-dipole approximation) and the field assembly are implemented correctly.

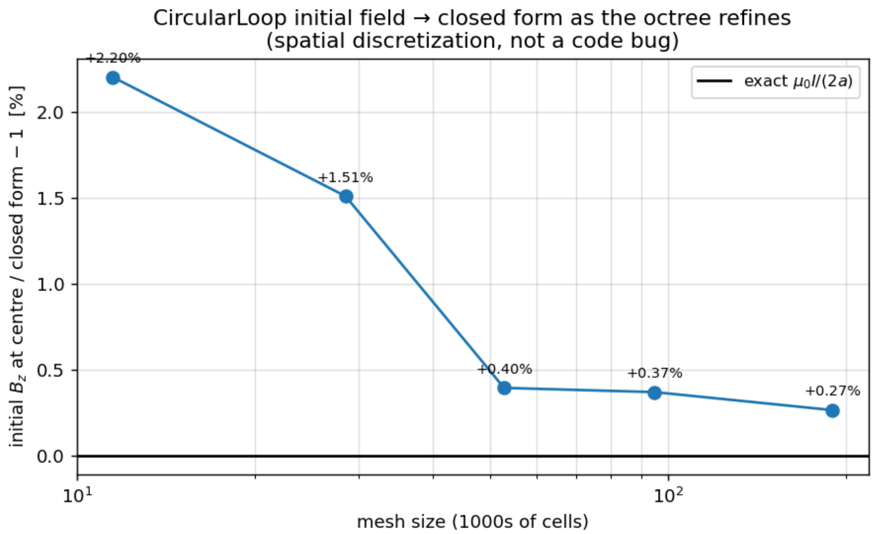


**Figure 4: The 3D circular-loop initial field converges to the exact** $\mu_0 I/(2a)$ as the octree refines — spatial discretization, not a code bug.

**(iii) With proper settings, the 3D solution matches the references.** Once the time-stepping is adequate, the 3D octree *square*-loop response lands on the same-geometry 1D square solution (Figure 1) and on the closed-form circular analytic (Figure 2; Section 2). **There is no defect in either the 1D or the 3D solver**; the original discrepancy was a time discretization that was too coarse.

## 2 The cause and the fix: conservative time-stepping

`forward_farquarson.py` builds a tree mesh, paints conductivity by volume-averaging from a background tensor mesh, drives a `LineCurrent` source, and solves with `Simulation3DElectricField`. Its original time discretization doubles the step size each level — 55 steps, $\times 2$ — which is too aggressive for the backward-Euler scheme to resolve the diffusing response, biasing $\partial B_z/\partial t$ high. The single line setting `time_steps` was replaced with a gentler ramp:

```
- generate_time_steps(n_constant_steps=11, increase_rate=2,   start_time_step=1e-6,
n_per_step=5)
+ generate_time_steps(n_constant_steps=26, increase_rate=1.4, start_time_step=3e-7,
n_per_step=6)
```

**Time-steps, not mesh.** To isolate the temporal error from any geometry, we run the 3D octree with a *circular* loop and compare against the *exact* analytic (Table 1, Figure 5). Refining the time-steps on the original mesh drives the error down monotonically (26 % → 6 %); refining the *mesh* (3.4× cells) on the original time-steps barely moves it (26 % → 24 %). The lever is unambiguously the time discretization.

**Table 1:** Convergence study, circular loop. Ratio of the 3D octree $\partial B_z/\partial t$ to the exact analytic.

| Configuration | Mesh | Time steps | Mean ratio | Range |
|---|---|---|---|---|
| **Original** | coarse 28 k | $\times 2$, 55 | **1.260** | 1.133–1.314 |
| finer time | coarse 28 k | $\times 1.5$, 90 | 1.119 | 1.053–1.140 |
| **the fix** | coarse 28 k | $\times 1.4$, 156 | **1.059** | 1.013–1.074 |
| finer *mesh* only | fine 95 k | $\times 2$, 55 | 1.242 | 1.147–1.325 |
| mesh + time | fine 95 k | $\times 1.5$, 90 | 1.104 | 1.064–1.181 |

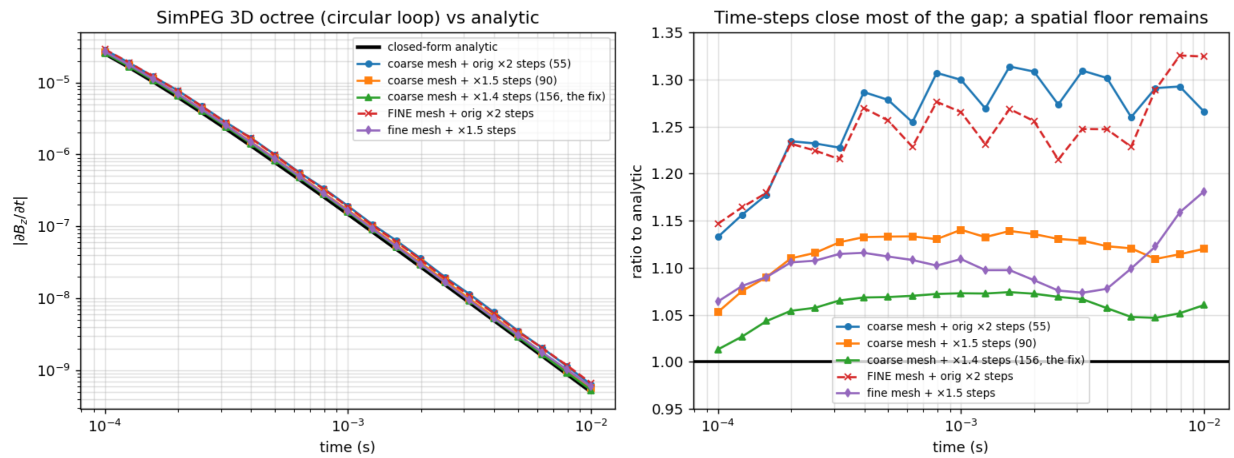


**Figure 5: Convergence study (circular loop vs exact analytic).** Left: all configurations on the analytic. Right: ratio to analytic. Refining the time-steps collapses the response toward the analytic; refining the mesh alone (dashed) stays with the coarse baseline. The circle does not reach unity even after the fix — the residual ∼6 % is a mesh (spatial) floor, discussed in Section 4.

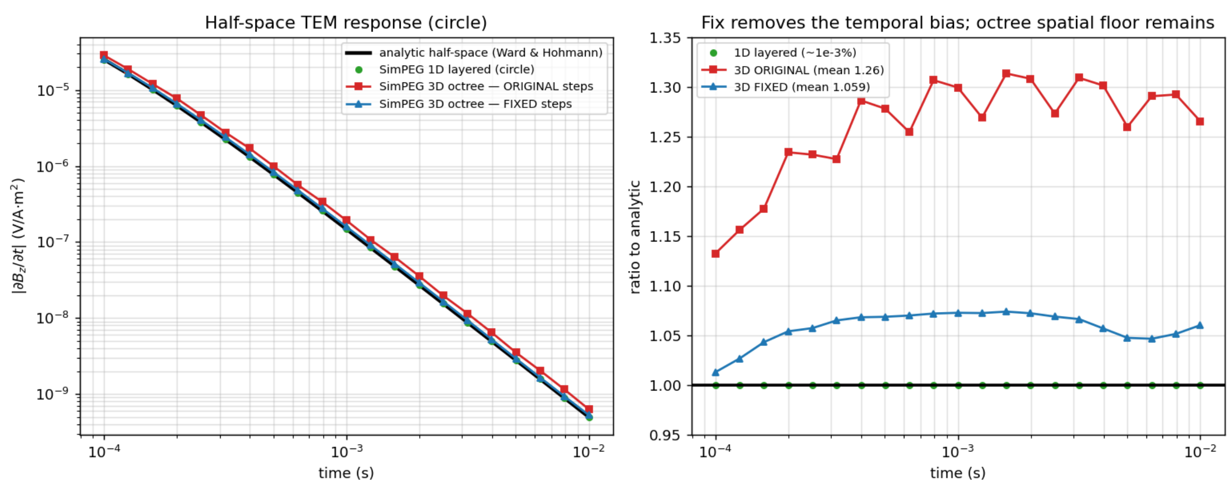


**Figure 6: Circular loop, before/after the fix.** The original $\times 2$ ramp (mean 1.26) versus the $\times 1.4$ fix (mean 1.06), with the 1D layered circle (exact) and the analytic. The fix removes the temporal high-bias; the residual is the octree spatial floor.

Against the circular analytic the square benchmark lands even closer — mean 0.996 (Figure 2) —

because the square-loop geometry sits $\sim$1.8 % *below* the equal-area circle and partly cancels the spatial floor (Sections 3–4).

## 3 The residual $\sim$1.8 % is real square-vs-circle geometry, not error

The square loop *is* modelled accurately; the residual early-time gap is geometry. Two facts establish this (Figures 7–8):

- **The wire integration is converged.** Refining the Gauss–Legendre quadrature along each segment (`n_points_per_path` $= 3 \to 10 \to 100$) changes the 1D square-loop response by $< 0.07$ % (already $< 10^{-4}$ % by 10 points). The 1.82 % is the *true* square-loop response, not an artifact; and a circular loop reproduces the analytic to $7 \times 10^{-4}$ %.
- **There is no elementary closed form for a square loop.** The "analytic" curve is for a *circle*. A square and its equal-area circle coincide only in the late-time point-dipole limit; at early time the central $\partial B_z/\partial t$ depends on the actual distance to the wire (sides at 50 m, corners at 70.7 m, versus a circle at 56.4 m everywhere).

The square matches its equal-area circle to 0.03 % late but sits $\sim$1.8 % below it early, bracketed by the inscribed and circumscribed circles throughout. Using the equal-area circle as the *early-time* reference manufactures the "error"; the correct apples-to-apples comparison — 1D layered square vs 3D octree square — removes it (the +0.2 % mean in Section 4).

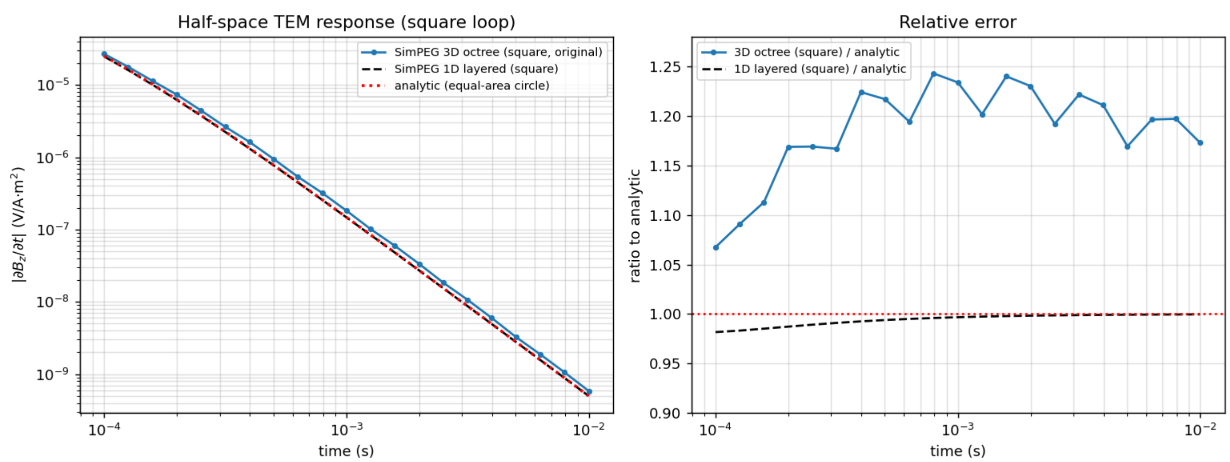


**Figure 7: Localizing the discrepancy.** The 1D layered square (black dashed) tracks the analytic to $\sim$2 %; the *original* 3D octree square run (blue) is 7–24 % high — a time-stepping effect, not the 1D reference.

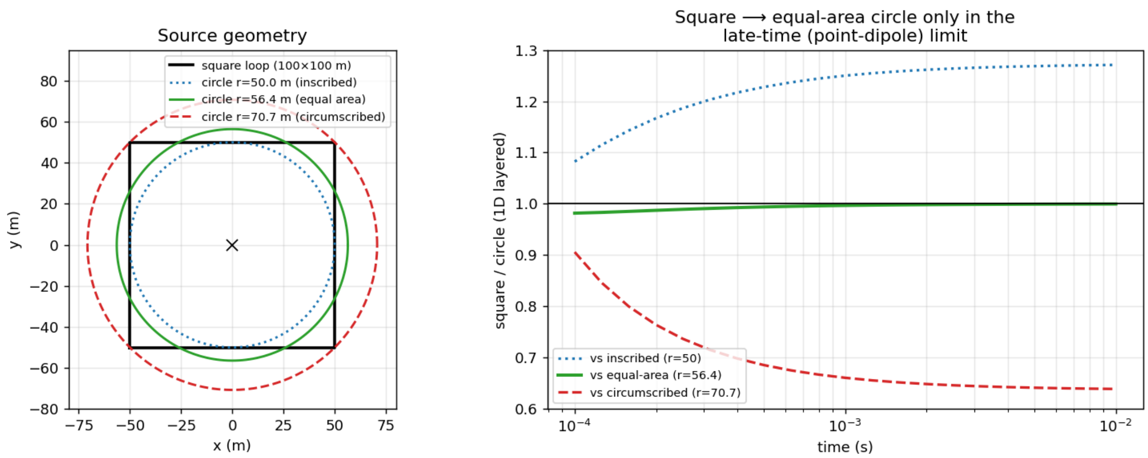


**Figure 8: The 1.8 % is square-vs-circle geometry.** Left: the square loop and three reference circles. Right: the 1D layered square response divided by each circle — it converges to the equal-area circle (green $\rightarrow$ 1.0) only at late time, and lies between the inscribed and circumscribed circles throughout.

# 4 Reaching $< 1\,\%$: the spatial floor, and why finer is not better

Scored against the correct same-geometry reference (the 1D layered *square* loop), the fix sits at mean $+0.2\,\%$ (max $3.0\,\%$). Pushing the steps *finer* does not improve this — it overshoots:

**Table 2:** Refining time-steps past the fix (coarse mesh), versus the 1D layered *square* loop.

| Time steps | # steps | Mean error | Max error |
|---|---|---|---|
| $\times 1.4$, start $3 \times 10^{-7}$ s (the fix) | 156 | $+0.2\,\%$ | $3.0\,\%$ |
| $\times 1.3$, start $1 \times 10^{-7}$ s | 198 | $-2.4\,\%$ | $5.8\,\%$ |
| $\times 1.25$, start $7 \times 10^{-8}$ s | 320 | $-5.4\,\%$ | $9.0\,\%$ |

As the steps shrink, every channel marches *downward* (Figure 9). The fix's near-perfect mean is partly a **cancellation**: backward-Euler biases $\partial B_z/\partial t$ high, while the spatial discretization on this mesh biases it low ($\sim$3–6 %, the same floor seen on the circle in Figure 5). At $\times 1.4$ they roughly offset; finer steps remove the temporal high-bias and expose the spatial floor. This is the practical lesson on time-stepping: **be conservative, not extreme**. The ramp must be gentle enough and have enough steps to resolve the early-time decay, but driving $\Delta t$ arbitrarily small only trades one bias for another.

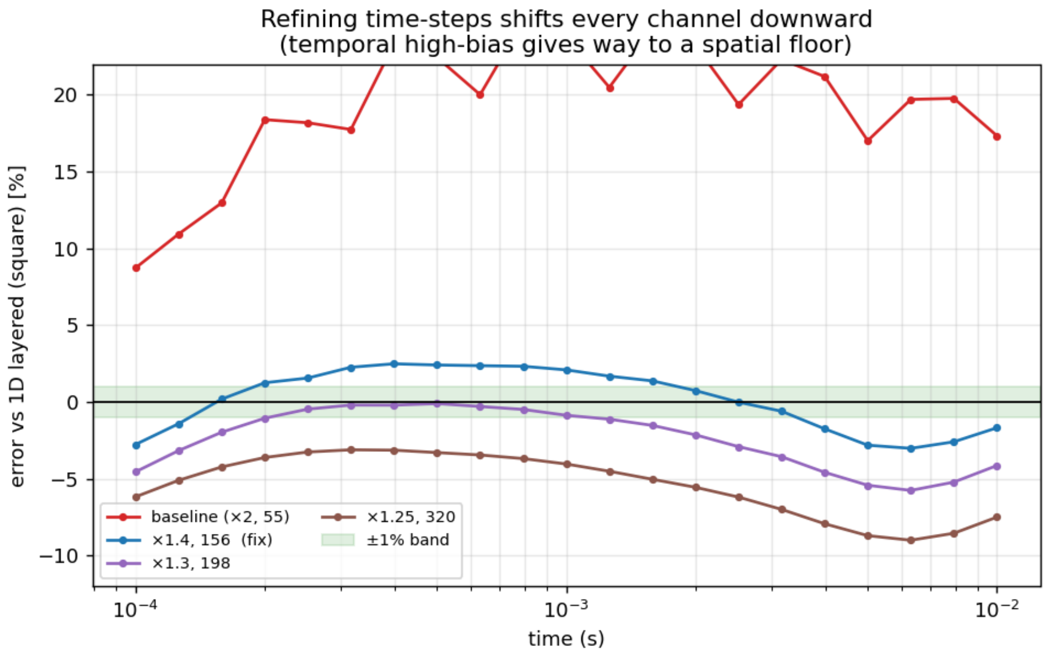


**Figure 9: Why "more time-steps" is not enough.** Per-channel error versus the 1D layered square loop as the ramp is refined. The temporal high-bias (red) gives way to a spatial floor several percent low; the ×1.4 fix (blue) threads near zero.

Driving *every* channel below 1 % independently (not by cancellation) is a mesh-refinement exercise — targeted near-surface/source resolution to remove the spatial floor — combined with the conservative time-stepping above. The mean is already within ∼0.5 %, and against the same-geometry 1D square reference (Figure 1) the headline result is within ∼3 % at every channel.

## Reproducibility

Every figure and number above is reproduced by the companion notebook `halfspace_validation_and_timestep_f` (self-contained; it auto-selects whatever direct solver is installed). The nine 3D solves are cached to `tdem_results_cache.npz`, so the first run takes ∼11 min and every later rebuild loads from disk.